

POWER BI

Interview Guide



Rajat Saxena
(SENIOR ANALYST)

Microsoft Power BI Interview Guide

- By Rajat Saxena

1. What is Power BI?

Answer: Power BI is a suite of Microsoft business analytics tools for connecting to data sources, modeling data, and building interactive reports and dashboards to share insights.

Explanation: It contains Power BI Desktop (authoring), Power BI Service (cloud sharing), Power BI Mobile, and gateways. It supports data ingestion, transformation, modeling, visuals, and distribution.

2. What are the main components of Power BI?

Answer: Power BI Desktop, Power BI Service (app.powerbi.com), Power BI Mobile apps, Power BI Report Server, and On-premises data gateway.

Explanation: Desktop is for creating reports; Service is for publishing, sharing, and scheduling refreshes; Gateway connects on-prem data; Report Server hosts on-prem reports.

3. What file formats does Power BI Desktop use?

Answer: .pbix for reports/datasets; .pbit for templates; .rdl for paginated reports (Power BI Report Server).

Explanation: .pbix contains data model, visuals, queries. .pbit saves structure without data.

4. How does Power BI differ from Excel?

Answer: Power BI focuses on scalable data modeling, DAX calculations, interactive visuals, and cloud sharing. Excel is flexible for ad-hoc analysis and cell-level formulas.

Explanation: Power BI handles larger models with VertiPaq engine, relationships, and centralized reports; Excel still useful for pivoting and personal analysis.

5. What is Power Query?

Answer: Power Query is the ETL (extract-transform-load) engine in Power BI Desktop used to connect to, clean, and transform data before loading to the model.

Explanation: It uses M language under the hood; transformations are recorded as steps and are repeatable.

6. What is DAX?

Answer: DAX (Data Analysis Expressions) is the formula language used in Power BI for creating measures, calculated columns, and calculated tables.

Explanation: DAX includes functions for aggregations, time intelligence, filters, and row context vs filter context concepts.

7. What is a measure vs a calculated column?

Answer: A measure is a dynamic aggregation calculated at query time, evaluated in filter context. A calculated column is computed row-by-row and stored in the model.

Explanation: Measures are preferred for aggregations (memory efficient). Calculated columns useful when you need categorical values for relationships or slicers.

8. What is the Power BI Service?

Answer: Power BI Service is the cloud platform (app.powerbi.com) for publishing reports, creating dashboards, sharing content, and scheduling data refresh.

Explanation: It includes workspaces, apps, datasets, dataflows, and supports row-level security and collaboration.

9. What are Power BI workspaces?

Answer: Workspaces are containers in Power BI Service for organizing dashboards, reports, datasets, and work with teammates. There are personal ("My workspace") and app workspaces.

Explanation: App workspaces allow permission control and packaging content into apps for distribution.

10. What is a Power BI dataflow?

Answer: Dataflows are Power Query-based ETL processes hosted in Power BI Service to create reusable, centralized entities (tables) stored in Azure Data Lake Storage Gen2 (or managed storage).

Explanation: Useful for standardizing transformations and sharing cleaned data between multiple reports.

11. What is M language?

Answer: M is the functional language used by Power Query to define data extraction and transformation steps.

Explanation: M queries are generated by UI actions; advanced transformations and parameterization use M directly.

12. How do you handle missing values in Power Query?

Answer: Use Replace Values, Remove Rows → Remove Blank Rows, or fill down/up. Also use conditional columns or custom M to handle blanks.

Explanation: Decide whether to remove, fill, or impute values depending on analysis needs.

13. How do you merge vs append queries?

Answer: Merge joins two queries horizontally (like SQL join). Append stacks queries vertically (union).

Explanation: Merge requires a key column and join type; Append requires matching column structure or maps columns.

14. What are query parameters and when to use them?

Answer: Parameters are dynamic values that can be referenced in queries (e.g., file path, date) to make queries reusable and configurable.

Explanation: Useful for environments, switching data sources, or pagination.

15. How to improve performance in Power Query?

Answer: Filter early, remove unused columns, disable background data preview, avoid row-by-row transformations, use native queries or query folding when possible.

Explanation: Query folding pushes transformations to the source (database) which is generally faster.

16. What is query folding?

Answer: Query folding is the ability of Power Query to translate applied transformations into source-side queries (e.g., SQL) to be executed on the data source.

Explanation: Transformations like filtering, joins fold; certain operations break folding (e.g., invoking custom functions).

17. How do you handle incremental refresh in Power BI?

Answer: Configure incremental refresh policies on a table in Power BI Desktop using rangeStart/rangeEnd parameters and then publish to Premium or Premium-per-user or Power BI Service with appropriate licensing.

Explanation: Incremental refresh processes only recent partitions to speed up refresh and scale to large datasets.

18. What are data types in Power Query?

Answer: Text, Whole Number, Decimal Number, Date, Time, DateTime, DateTimeZone, Duration, True/False, Binary.

Explanation: Correct types are important for subsequent transformations and modeling.

19. How to pivot and unpivot columns?

Answer: Use Transform → Pivot Column or Unpivot Columns in Power Query to reshape data from long to wide or vice versa.

Explanation: Unpivot is commonly used to convert cross-tab data into tidy format for modeling.

20. What is a staging query and why use it?

Answer: Staging queries are intermediate queries used to shape raw data (cleanup) and act as a base for other queries. Marking as “Disable load” prevents them from being loaded into the model.

Explanation: Helps modularize ETL and improve performance.

21. How do you combine files from a folder?

Answer: Use Get Data → Folder, then Combine Files which creates a sample query and applies transformations to all files in that folder.

Explanation: Good for periodic file ingestion with consistent schema.

22. What is the difference between "Close & Apply" and "Apply" in Power Query?

Answer: Close & Apply saves query changes and applies them to the model and refreshes the dataset; Apply loads changes but keeps Query Editor open.

Explanation: Use Apply to test incremental steps without closing the editor.

23. How to handle duplicate rows?

Answer: Use Remove Rows → Remove Duplicates in Power Query, specifying columns for deduplication.

Explanation: Often done after data consolidation to ensure unique records.

24. How can you parameterize data source connections for different environments?

Answer: Create parameters for server, database, file path, etc., and reference them in your queries. Use templates (.pbit) or manage parameters in Power BI Service.

Explanation: Eases deployment across dev/test/prod.

25. How do you extract values from JSON or nested structures?

Answer: Use Power Query's JSON.Document, expand records and lists in the UI to drill into nested objects and convert them to table columns.

Explanation: Power Query provides expand buttons and functions to normalize semi-structured data.

26. What is star schema and why is it important?

Answer: Star schema organizes data into fact tables (measures) and dimension tables (attributes) with relationships; it simplifies queries and improves DAX performance.

Explanation: Reduces complexity and prevents ambiguous relationships; favored over highly normalized schemas for analytics.

27. What are relationships in Power BI?

Answer: Relationships connect tables on keys (one-to-many, many-to-many, one-to-one) and control how filters flow across tables.

Explanation: Cardinality and cross-filter direction affect how slicers and measures behave.

28. Explain cardinality types.

Answer: Common cardinalities: One-to-many, Many-to-one, One-to-one, Many-to-many.

Explanation: One-to-many is typical (dimension to fact). Many-to-many can be modeled using bridging tables or composite models in recent versions.

29. What is cross-filter direction?

Answer: Cross-filter direction (Single or Both) determines how filters propagate between related tables. Single flows from the one side to the many; Both allows bi-directional filtering.

Explanation: Both can simplify modeling but may introduce ambiguity and performance issues.

30. What is a calculated table? When to use it?

Answer: A calculated table is created with a DAX expression at model processing time. Use for static derived tables, summary tables, or bridging tables.

Explanation: Useful for scenario tables or precomputed groupings.

31. What are perspectives?

Answer: Perspectives are curated views of a model that limit which tables/columns/measures are visible to report consumers.

Explanation: Helps simplify the model for specific roles in Power BI Desktop or Tabular Editor.

32. How do you handle slowly changing dimensions in Power BI?

Answer: Approaches: model Type 1 (overwrite), Type 2 (history via effective-dated rows), or use snapshot tables and change data capture upstream. Implement Type 2 via dataflow queries or in the source.

Explanation: Power BI itself is primarily read-only for modeling; often best to handle SCD in ETL.

33. What is row-level security (RLS)?

Answer: RLS restricts data a user can see by applying filters to tables based on roles in the model. Can be static (role-based filters) or dynamic using USERNAME() or USERPRINCIPALNAME().

Explanation: Configure roles in Desktop, then assign users in Service or Azure AD when publishing apps.

34. What is a surrogate key and why use it?

Answer: A surrogate key is an artificial unique identifier (integer) for records used instead of natural keys to maintain stable joins and performance.

Explanation: Helps when natural keys are composite or change over time; improves storage and indexing.

35. How to optimize your data model for performance?

Answer: Remove unused columns, use star schema, reduce cardinality (bucket values), use measures instead of calculated columns, disable unnecessary relationships, and reduce data types (use integer when possible).

Explanation: VertiPaq compression benefits from repeated values and lower cardinality.

36. What is a composite model?

Answer: Composite models combine import-mode tables and DirectQuery tables (or multiple DirectQuery sources) in the same model.

Explanation: Provides flexibility to mix real-time querying with in-memory tables; consider performance and limitations.

37. When to use DirectQuery vs Import mode?

Answer: Use Import for speed and complex analytics; DirectQuery for very large datasets or when data must be real-time or governed at source.

Explanation: Import caches data in-memory (fast), DirectQuery queries the source on demand with restrictions and possible latency.

38. What is dual storage mode?

Answer: Dual mode allows a table to behave as both Import and DirectQuery: it can serve cached values for aggregations while also supporting DirectQuery for detail queries.

Explanation: Useful for aggregation tables to speed queries.

39. How do you model many-to-many relationships?

Answer: Use a bridge (junction) table with unique keys, or use native many-to-many relationships (table marked as many-to-many) with careful filter considerations. Prefer explicit bridging dimension for clarity.

Explanation: Many-to-many can create ambiguous filter propagation and performance issues if not designed well.

40. What is the VertiPaq engine?

Answer: VertiPaq is Power BI's in-memory columnar storage and compression engine that stores and queries imported data efficiently.

Explanation: It compresses columns based on patterns and encodings, enabling fast analytical queries.

41. What is the difference between SUM and SUMX?

Answer: SUM aggregates a single column directly; SUMX iterates a table row-by-row evaluating an expression for each row then sums results.

Explanation: Use SUMX when you need to compute per-row expressions (e.g., quantity * price) before aggregating.

42.Explain filter context vs row context.

Answer: Row context iterates over table rows (calculated columns, iterators). Filter context is the set of filters applied to a calculation (slicers, report filters, relationships).

Explanation: Functions like CALCULATE transform filter context; to access row context inside measures you often need to use aggregation functions.

43.What does CALCULATE do?

Answer: CALCULATE modifies the current filter context by applying or removing filters, and then evaluates an expression in that modified context.

Explanation: Essential for advanced DAX like time-intelligence or conditional aggregations.

44.How to write a measure for Year-to-Date sales?

Answer: Use TOTALYTD([Sales], 'Date'[Date]) or CALCULATE([Sales], DATESYTD('Date'[Date])).

Explanation: Requires a proper date table marked as Date table and a contiguous date column.

45.What is an evaluation context transition?

Answer: When a row context is converted into an equivalent filter context—this happens when CALCULATE is used inside a row context.

Explanation: Important to reason about how nested calculations behave.

46. How to create a running total in DAX?

Answer: Use CALCULATE with FILTER over dates up to current date, e.g.,
`CALCULATE([TotalSales], FILTER(ALL('Date'), 'Date'[Date] <= MAX('Date'[Date])))` or
use TOTALYTD/TOTALMTD.

Explanation: Use proper ALL/ALLSELECTED to control slicer behavior.

47. What is ALL function used for?

Answer: ALL removes filters from specified columns or tables, returning an unfiltered table for calculations.

Explanation: Useful for percentages of total and ignoring slicer filters where appropriate.

48. What is RELATED function?

Answer: RELATED retrieves a value from a related table in a one-to-many relationship; it uses the relationship to fetch the related row value.

Explanation: Common in calculated columns when you need lookup-like behavior.

49. How to handle divide by zero in DAX?

Answer: Use DIVIDE(numerator, denominator, alternateResult) which safely handles zero or use IF(denominator = 0, BLANK(), numerator/denominator).

Explanation: DIVIDE is recommended because it also returns blank when numerator is blank.

50. What is EARLIER? Why might you avoid it?

Answer: EARLIER references a value from an earlier row context in nested row contexts; often complex and replaced by variables or SUMX for clarity.

Explanation: EARLIER can be confusing; use variables (VAR) and nested iterators instead.

51. Explain the use of variables (VAR) in DAX.

Answer: VAR stores intermediate results in a measure or calculated column to improve readability, reuse values, and sometimes performance.

Explanation: Use RETURN to output the final result; variables are evaluated once per evaluation context.

52. What is DISTINCTCOUNT? How does it affect performance?

Answer: DISTINCTCOUNT counts unique values in a column. It can be more expensive than COUNTROWS on a smaller table because it needs to de-duplicate values.

Explanation: Consider pre-aggregating or using optimized model design to reduce distinct counts.

53. How to calculate percentage growth between current and previous period?

Answer: Use measures with time intelligence: $([ThisPeriod] - [PreviousPeriod]) / [PreviousPeriod]$ or use DIVIDE for safety. PreviousPeriod often computed via DATEADD or SAMEPERIODLASTYEAR.

Explanation: Requires correctly marked date table.

54. What is ISFILTERED and HASONEVALUE?

Answer: ISFILTERED checks if a column has any filter applied. HASONEVALUE returns true if exactly one distinct value is in filter context for a column.

Explanation: Useful for dynamic titles or conditional measures.

55. How to convert text to number in DAX?

Answer: Use VALUE(text) or INT/FIXED functions for some conversions.

Explanation: Ensure data types are correct; Power Query is often better place to fix types.

56. Explain context transition with an example.

Answer: Example: In a calculated column, using CALCULATE([Measure]) will convert the row context of the current row into filters for the measure. This changes how the measure is computed.

Explanation: Understanding this helps avoid unexpected results.

57. What is USERPRINCIPALNAME()?

Answer: A DAX function returning the current user's principal name (email) used for dynamic RLS and personalization.

Explanation: Combine with lookup tables to implement dynamic RLS.

58. What's the difference between FILTER and CALCULATETABLE?

Answer: FILTER returns a table after evaluating a Boolean expression row-by-row; CALCULATETABLE modifies filter context and returns a table — it can accept filter expressions and perform context transition.

Explanation: Use FILTER when needing row-by-row logic; CALCULATETABLE when altering filters in a DAX measure.

59. What are iterator functions in DAX? Name some.

Answer: Iterators loop over a table evaluating an expression per row, e.g., SUMX, AVERAGEX, COUNTX, MINX, MAXX.

Explanation: They create row context; often used for per-row calculations aggregated afterward.

60. How to build a dynamic title for a visual?

Answer: Create a measure returning a text string and use it in the visual title via Title fx (conditional formatting).

Explanation: Use IF and SELECTEDVALUE or HASONEVALUE to display selected slicer values.*

61. How to write a measure for running average?

Answer: Use AVERAGEX over a filtered date range: AVERAGEX(DATESBETWEEN('Date'[Date], START, MAX('Date'[Date])), [DailyMeasure]) or use moving average patterns using TOPN or FILTER.

Explanation: Choose right window size and manage empty dates by using a continuous date table.

62. Explain how to use ALLSELECTED.

Answer: ALLSELECTED retains filters applied outside the visual (e.g., outer slicers) while removing filters inside the visual, useful for percent-of-selected calculations.

Explanation: Useful for dynamic percentages based on user selections.

63. How to handle sparse date coverage in time intelligence?

Answer: Use a complete calendar/date table marked as Date table with contiguous dates; join to fact table and use measures referencing date table to ensure continuous periods.

Explanation: Missing dates in the date table disrupt time-intel functions.

64. What is calculation group (in Tabular Editor) and how does it help?

Answer: Calculation groups are a Tabular model feature that let you define reusable calculation expressions (e.g., time-intel like YTD, QoQ) applied to measures to reduce measure duplication and simplify models.

Explanation: They can greatly reduce measure bloat and be applied across measures; requires care with formatting and interactions.

65. How to implement dynamic segmentation (e.g., RFM) in Power BI?

Answer: Compute R, F, M metrics using measures or calculated columns, then create buckets via DAX (SWITCH/IF) or use calculated tables to define segments. Visualize segments with slicers.

Explanation: Use measures if segmentation needs to be dynamic; calculated columns when segments are stable and used for relationships.

66. How to compute moving median?

Answer: Create a measure that collects values over a window using FILTER and then uses MEDIANX over the result: MEDIANX(VALUES('Date'[Date]) filtered by window, [ValueMeasure]).

Explanation: MEDIAN is not additive; using MEDIANX gives per-row median across selected window.

67. What are virtual relationships?

Answer: Virtual relationships are created by functions like TREATAS or CROSSFILTER/CALCULATETABLE that apply filters across tables without an explicit model relationship.

Explanation: Useful for dynamic joins or when source relationships are impractical.

68. How does TREATAS work? Give an example.

Answer: TREATAS applies values from a table expression as filters on columns of another table.

Example: `CALCULATE([Sales], TREATAS(VALUES('SelectedProducts'[ProductKey]), 'Product'[ProductKey]))` applies product selection as a filter on Product.

Explanation: Powerful for slicer-like behavior from disconnected tables.

69. How to handle Many-to-Many in DAX measures?

Answer: Use measures that create explicit filter tables via VALUES or DISTINCT and then use TREATAS or CROSSFILTER to apply filters, or use bridge tables with measures referencing them.

Explanation: Keep logic explicit and avoid bi-directional filters that cause ambiguity.

70. How to build a contribution to total measure?

Answer: `Contribution % = DIVIDE([ThisMeasure], CALCULATE([ThisMeasure], ALL('Dimension')))` to calculate the share of total across dimension values.

Explanation: Choose ALL or ALLSELECTED depending on whether you want to ignore slicers.

71. How to write a measure to show Top N items dynamically?

Answer: Use RANKX in combination with FILTER and TOPN. Example: Show items where `RANKX(ALL('Product'), [Sales]) <= [ParameterTopN]`.

Explanation: Use parameters (What-if) to let users change N.

72. How to compute cohort retention using DAX? (high-level)

Answer: Create cohort (e.g., first purchase month) as a calculated column, then build measures that count active users in subsequent periods grouped by cohort using FILTER over dates/cohort and DIVIDE to compute retention percentage.

Explanation: Requires proper date relationships and possibly user-level attribution.

73. Explain the use of SUMMARIZE vs SUMMARIZECOLUMNS.

Answer: SUMMARIZE creates a table grouped by columns and can include expressions; SUMMARIZECOLUMNS is optimized for tabular and handles filter context better and is preferred for performance.

Explanation: SUMMARIZECOLUMNS avoids unexpected context transition behavior and is recommended for new models.

74. What is ISINSCOPE and how is it used?

Answer: ISINSCOPE checks whether a column is in the current drilldown scope (visual row context). Useful for dynamic measures that behave differently at different grouping levels.

Explanation: Helps create measures that vary by hierarchy level.

75. How to debug DAX performance?

Answer: Use Tools: DAX Studio (Server Timings / Query Plan), VertiPaq Analyzer, SQL Profiler for DirectQuery, and use variables, reduce row context, and simplify filters.

Explanation: Profiling helps pinpoint slow parts like expensive iterators or queries not folding to source.

76. How to write efficient measure for distinct count with filters?

Answer: Use DISTINCTCOUNT on a prefiltered table: `CALCULATE(DISTINCTCOUNT('Table'[Key]), FILTER( ALL('Table'), conditions ) )` and prefer indexed keys and star schema.

Explanation: Sometimes using COUNTROWS(VALUES(...)) or creating aggregated tables can be faster.

77. How to implement semi-additive measures (e.g., account balance)?

Answer: Use functions like LASTNONBLANK or calculate balance at period end: `CALCULATE( SUM('Fact'[Balance]), LASTDATE('Date'[Date]) )` or more complex time-aware logic.

Explanation: Semi-additive measures cannot be summed across time; need period-aware aggregation.

78. Explain query context in DAX and give an example.

Answer: Query context is established when a query requests values (e.g., visuals request a measure across rows), which defines filter contexts for each cell. Example: A table visual requesting Sales by Product creates filter context per product.

Explanation: Measures evaluate per-cell within that context.

79. How to implement dynamic measure switching?

Answer: Use disconnected tables with measure names and SELECTEDVALUE to choose which measure to show via SWITCH or use calculation groups in Tabular Editor for cleaner implementation.

Explanation: Useful for building user-controlled metric pickers.

80. How to compute Year-over-Year growth for a rolling window?

Answer: Use moving window measures with DATEADD or SAMEPERIODLASTYEAR combined with CALCULATE to fetch previous year value and compute percentage change over the aggregated window.

Explanation: Careful with granularity and alignment of dates.

81. What makes an effective Power BI visual/dashboard?

Answer: Clarity, relevance, concise layout, right chart types, proper use of color and labels, interactive filters, and clear call-to-action. Avoid clutter and emphasize key KPIs.

Explanation: Use appropriate visuals (bar/line for trends), ensure accessibility and consistent formatting.

82. How to create drill-through and drill-down?

Answer: Drill-down uses hierarchies on visuals (expand/collapse); Drill-through uses a target page with a drill-through filter field allowing users to right-click to navigate to details for one selection.

Explanation: Drill-through is for detail pages; drill-down navigates within a visual hierarchy.

83. How to create bookmarks and what are they for?

Answer: Bookmarks capture the report state (filters, slicers, visuals) and are used for storytelling, navigation, and resetting views.

Explanation: Combine with buttons for custom navigation experiences.

84. What is conditional formatting? Give examples.

Answer: Conditional formatting adjusts colors, fonts, or icons based on data (e.g., color scales for KPI, data bars, icons). Can be based on measures to apply business rules.

Explanation: Use formatting to call attention but avoid overuse.

85. When to use slicers vs filters?

Answer: Use slicers for visual, page-level filtering by users; filters pane is better for persistent, less intrusive filters. Slicers are interactive and visible on the canvas.

Explanation: For single selection or UX-driven filtering, use slicers; for hidden or admin filters use filter pane.

86. How to design for mobile in Power BI?

Answer: Create a Mobile Layout in Power BI Desktop or configure visuals for phone layout; prioritize essential visuals with simplified navigation and legible text.

Explanation: Mobile view allows arranging visuals specifically for small screens.

87. What is Q&A visual? How to improve results?

Answer: Q&A allows natural language queries over your dataset. Improve by defining synonyms, phrasings, and having clear field display names and a good semantic model.

Explanation: Add featured Q&A questions for guidance.

88. How to export Power BI visuals for presentations?

Answer: Export to PDF, PowerPoint (Export > PowerPoint), or export data to CSV/Excel. For higher fidelity, use Paginated Reports for pixel-perfect exports.

Explanation: PowerPoint export creates images of report pages; paginated reports handle multi-page, tabular exports.

89. What is a custom visual and how to add one?

Answer: Custom visuals are visuals developed using Power BI Visuals SDK or from AppSource. Import via "Get more visuals" or "Import a visual from file".

Explanation: Use well-supported visuals and check security and performance implications.

90. How to make visuals accessible?

Answer: Provide alt text, readable fonts, high-contrast colors, consistent legends, avoid color-only distinctions, and ensure keyboard navigation where possible.

Explanation: Improves usability for users with disabilities and meets accessibility standards.

91. How to schedule data refresh in Power BI Service?

Answer: Configure dataset scheduled refresh in Power BI Service under dataset settings; ensure gateway is configured for on-prem sources and credentials are set. Refresh frequency depends on license (Pro vs Premium).

Explanation: Pro users can schedule up to 8 daily refreshes (as of last known limits); Premium allows more frequent refreshes and larger models.

92. What is On-premises data gateway?

Answer: The gateway bridges secure data transfer between on-premise data sources and Power BI Service, enabling scheduled refresh and DirectQuery to on-prem sources.

Explanation: Two modes: personal (single-user) and enterprise (multi-user/high-availability).

93. What are dataset size limits and how to handle large models?

Answer: Import models have size limits depending on license (Pro ~1 GB, Premium higher). For large models use incremental refresh, aggregations, composite models,

DirectQuery, or Premium capacity.

Explanation: Also optimize model by removing columns and reducing cardinality.

94. How to monitor and optimize report performance?

Answer: Use Performance Analyzer in Power BI Desktop, DAX Studio for queries, optimize visuals (limit complex custom visuals), reduce visuals per page, and optimize DAX and model.

Explanation: Performance Analyzer shows visual load times and DAX queries to target bottlenecks.

95. What is Power BI App and how to share reports via it?

Answer: An app packages related dashboards/reports/datasets from a workspace into a consumable, permissioned package for end-users. Publish an app from workspace and assign access.

Explanation: Apps provide controlled distribution and versioning.

96. How to implement row-level security (RLS) in published reports?

Answer: Define RLS roles in Power BI Desktop (DAX filters), publish dataset, then assign users/groups to roles in the Power BI Service or via Azure AD. For dynamic RLS use USERPRINCIPALNAME() with mapping table.

Explanation: Test roles in Desktop and Service; apps must be republished after role changes for users to see correct access.

97. What is the difference between Power BI Pro and Premium?

Answer: Pro is per-user licensing for authoring and sharing with limits; Premium offers dedicated capacity, larger dataset sizes, more refreshes, paginated reports, and advanced features. Premium per user (PPU) adds Premium features per user.

Explanation: Choose based on scale, performance, and enterprise needs.

98. How to set up data lineage and impact analysis?

Answer: Use Power BI lineage view in Service to visualize relationships between datasets, reports, dataflows, and gateways; manage dependencies and impact before changes.

Explanation: Helps governance and troubleshooting.

99. How to secure data in Power BI (best practices)?

Answer: Use RLS, secure workspaces, Azure AD groups, data classification, sensitivity labels, enforce TLS, restrict downloads/exports, and govern tenant-level settings in Admin Portal.

Explanation: Combine model-level and service-level controls; train users on data handling.

100. Practical interview question — Walk me through building a sales analysis report end-to-end.

Answer (concise steps):

- Understand requirements and KPIs.
- Identify and connect to data sources (sales, products, customers, calendar).
- Use Power Query to clean, transform, and shape data (dates, types, missing values, dedupe).
- Build a star schema: fact table (sales) and dimension tables (date, product, customer).
- Create measures (Total Sales, Units, Gross Margin, YoY growth) using DAX with variables and time-intel.
- Implement RLS for sensitive sales regions if required.

- Design visuals: KPI cards, trend lines, bar charts, maps, table with drill-through, and slicers. Add bookmarks for storytelling.
- Optimize: remove unused fields, reduce cardinality, enable incremental refresh, and test performance.
- Publish to Power BI Service, set up scheduled refresh and workspace/app sharing, and monitor usage.

Explanation: Shows end-to-end capability; in interviews expand on choices (import vs DirectQuery, incremental refresh, stakeholder communication).

Thank You for Reading

Thank you for taking the time to go through *Power BI Interview Guide*.

I truly hope this guide helps you strengthen your Power BI concepts, build confidence, and perform better in interviews.

This guide was created with one clear goal — **to genuinely help job seekers prepare smarter and land the right opportunities**. If even a small part of this book adds value to your preparation, it means a lot to me.

Your learning journey doesn't have to stop here. I regularly share **interview questions, job updates, preparation tips, and career guidance** on my platforms. Feel free to connect and stay in touch:

 [LinkedIn](#)

 [WhatsApp Channel](#)

 [Telegram Channel](#)

If you found this guide helpful, do share it with fellow job seekers who might benefit from it.

Wishing you **all the very best for your Power BI interviews and career growth** 🚀

Warm regards,

[Rajat Saxena](#)